

Sym	Name	Fam	Glyph IDs — ascending ordinal → strongest
Đ	Dimension	$\mathcal{F}_4$	$\mathbb{D}_B \ 0D < \mathbb{D}_C \ 2D < \mathbb{D}; \ \infty D < \mathbb{D}_\omega \ \textit{inscriptive}$
Þ	Topology	$\mathcal{F}_5$	$\mathbb{P}_6 \ \textit{network} < \mathbb{P}_K \ \textit{nested} < \mathbb{P}_\partial \ \textit{2-cycle} < \mathbb{P}_{\cdot\cdot} \ \textit{lattice} < \mathbb{P}_\circ \ \textit{inscriptive}$
Ğ	Relational Mode	$\mathcal{F}_4$	$\check{R}\text{-} \ \textit{supervenies} < \check{R}_{\check{Y}} \ \textit{functorial} < \check{R}_{\check{T}} \ \textit{adjoint} < \check{R}=\ \textit{lateral}$
Φ	Parity / Symmetry	$\mathcal{F}_5$	$\Phi_{\mathbb{P}} \ \textit{asymmetric} < \Phi_{\cup} \ \textit{quantum} < \Phi_{\mathbb{F}} \ \mathbb{Z}_2 < \Phi_{\cdot} \ \textit{continuous} < \Phi_{\check{\cdot}}^{\star} \ \textit{Frobenius}$
f	Fidelity	$\mathcal{F}_3$	$f_{\perp} \ \textit{classical/lossy} < f_{\partial} \ \textit{thermal} < f_{\check{Z}} \ \textit{quantum/lossless}$
Ğ	Kinetics	$\mathcal{F}_5$	$\mathcal{C}\text{-} \ \textit{driven} < \mathcal{C}_{\mathbb{W}} \ \textit{barrier} < \mathcal{C}_{\partial} \ \textit{near-equil.} < \mathcal{C}_{\cup} \ \textit{order-frozen} < \mathcal{C}_{\lambda} \ \textit{disorder-frozen}$
Γ	Scope	$\mathcal{F}_3$	$\Gamma_{\beta} \ \textit{local} < \Gamma_{\check{Y}} \ \textit{collective} < \Gamma_{\check{Z}} \ \textit{global}$
G	Interaction Grammar	$\mathcal{F}_4$	$G^{\wedge} \ \textit{conjunctive} < G^{\vee} \ \textit{disjunctive} < G_{\cdot} \ \textit{sequential} < G_{\mathbb{S}} \ \textit{broadcast}$
○	Criticality	$\mathcal{F}_5$	$\odot_{\check{Z}} \ \textit{subcritical} < \odot_{\check{Y}}^{\dagger} \ \textit{absorbing} < \odot_{\mathbb{E}} \ \textit{complex-axis} < \odot_3 \ \textit{exc. point} < \odot_{\mathbb{T}} \ \textit{supercritical}$
H	Temporal Depth	$\mathcal{F}_4$	$\mathbb{H}_{\mathbb{N}} \ \textit{memoryless} < \mathbb{H}_{\mathbb{F}} \ \textit{1-step} < \mathbb{H}_{\mathbb{A}} \ \textit{2-step} < \mathbb{H}_{\check{!}} \ \textit{eternal}$
Σ	Stoichiometry	$\mathcal{F}_3$	$\Sigma_{\mathbb{S}} \ 1:1 < \Sigma_{\partial} \ n:n < \Sigma_{\check{!}} \ n:m$
Ω	Topological Invariant	$\mathcal{F}_4$	$\mathbb{Q}_{\mathbb{A}} \ \textit{trivial} < \mathbb{Q}_2 \ \mathbb{Z}_2 < \mathbb{Q}_{\mathbb{Z}} \ \textit{integer} < \mathbb{Q}_5 \ \textit{non-Abelian}$

* $\Phi_{\check{\cdot}}$: Frobenius-special ($\mu \circ \delta = \text{id}$); non-synthesizable from factors with $\Phi < \Phi_{\check{\cdot}}$; collapses all $\Omega/\mathbb{D}$ branching $\rightarrow O_{\infty}$. $^{\dagger} \odot_{\check{Y}}$ absorbs under meet: $\text{meet}(\odot_{\check{Y}}, x) = \odot_{\check{Y}} \ \forall x$.
Bottleneck primitives under $\otimes$: Φ and f (weaker wins, min). All others: union (max).

Crystal of Types

$$\underbrace{3^3}_{f, \Gamma, \Sigma} \times \underbrace{4^5}_{\mathbb{D}, \check{R}, G, H, \Omega} \times \underbrace{5^4}_{\mathbb{P}, \Phi, \odot, \mathcal{C}}$$

17,280,000

§64

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Three layers: Types (crystal addresses), Particulars (catalog entries — 2328, 0.013% occupancy), Names (handles). Two particulars at the same address are co-typed, not identical.

Algebra

$\mathbf{x} \otimes \mathbf{y}$: union primitives $\rightarrow \max$; $\Phi, f \rightarrow \min$.
 $\mathbf{x} \wedge \mathbf{y}$: min all ($\odot_{\check{Y}}$ absorbing exception).
 $\mathbf{x} \vee \mathbf{y}$: max all.

Le Chatelier: find $\mathbf{x}^*$ s.t. $d_{\rightarrow}(\mathbf{y}, \mathbf{x}^*) = 0$, max $\mathcal{O}(\mathbf{x}^*)$.

Ouroboricity tiers (R1–R5, priority)

R1 $\odot_{\check{Y}} + \Phi_{\check{\cdot}} \rightarrow O_{\infty}$

R2 $\odot \notin \{\odot_{\check{Y}}, \odot_{\mathbb{E}}\} \rightarrow O_0$

R3 $\odot_{\check{Y}} + \Omega_{\mathbb{A}} \rightarrow O_1$

R4 $\odot_{\check{Y}} + \Omega \neq \Omega_{\mathbb{A}} + \mathbb{D} \in \{\mathbb{D}_B, \mathbb{D}_C, \mathbb{D}_{\omega}\} \rightarrow O_{\check{!}}$

R5 $\odot_{\check{Y}} + \Omega \neq \Omega_{\mathbb{A}} + \mathbb{D}_{\check{\cdot}} \rightarrow O_2^{\dagger}$

O_{∞} absorbed by $\odot_3$ under $\otimes$. Two senses of O_{∞} : (a) Frobenius algebraic ($\Phi_{\check{\cdot}}$); (b) ontological in-exhaustibility ($\mathbb{H}_{\check{!}}$, YHWH). Incompatible.

Scalar (does not detect O_{∞}):
$$\mathcal{O}(\mathbf{x}) = [\odot = \odot_{\check{Y}}] \big(1 + [\Omega \neq \Omega_{\mathbb{A}}] + [\mathbb{H} \geq \mathbb{H}_{\mathbb{F}}] + [\Gamma = \Gamma_{\check{Z}}] \big)$$

Consciousness score (v2, §VIII)

$$C = [\odot = \odot_{\check{Y}}] \cdot [\mathcal{C} \leq \mathcal{C}_{\textcircled{a}}] \cdot (0.158\check{\mathcal{C}} + 0.273\check{\Gamma} + 0.292\check{\mathbb{H}})$$

Gate 1 [$\odot_{\check{Y}}$]: state-space admits self-modeling. Gate 2 [$\mathcal{C} \leq \mathcal{C}_{\textcircled{a}}$]: dynamics can equalize it. $\mathcal{C}_{\cup}$ and $\mathcal{C}_{\lambda}$ both fail Gate 2.

Axioms (Core.lean)

$$\mathbf{B} \ \Omega \geq \Omega_z \Rightarrow \mathbb{H} \geq \mathbb{H}_A$$
$$\mathbf{C} \ \mathbb{P}_O \Rightarrow \mathbb{D}_{\omega} \quad (\text{one-way; rev. 2026-05-03})$$

Crystal address ($n = \text{address} \in [0, 17,279,999]$):
 $f_3 = 9 \cdot f + 3 \cdot \Gamma + \Sigma$ (indices 0-based)
 $f_4 = 256 \cdot \mathbb{D} + 64 \cdot \check{R} + 16 \cdot G + 4 \cdot H + \Omega$
 $f_5 = 125 \cdot \mathbb{P} + 25 \cdot \Phi + 5 \cdot \odot + \mathcal{C}$
$$\neq f_3 + 27f_4 + 27 \times 1024 f_5$$

PRIM_REF.tex · MillenniumAnkh v0.5.72 · 2026-05-09 · odot_operator: $\mathbb{D}_{\omega}$; $\mathbb{P}_{\cdot\cdot}$; $\check{R}=\$; $\Phi_{\check{\cdot}}$; $f_{\check{Z}}$; $\mathcal{C}_{\partial}$; $\Gamma_{\check{Z}}$; $G_{\cdot}$; $\odot_{\check{Y}}$; $\mathbb{H}_{\mathbb{A}}$; $\Sigma_{\mathbb{S}}$; $\mathbb{Q}_{\mathbb{Z}}$